

VEHICLE POWER ADAPTOR

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Laptops come with a mains adaptor as the standard accessory. However, if you want to power your laptop while traveling in

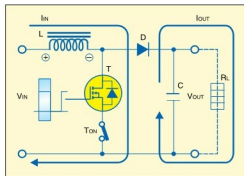


Fig. 1: Operating principle of step-up converter during switch-on phase

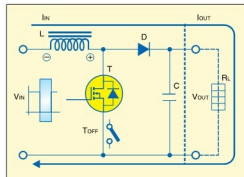


Fig. 2: Operating principle of step-up converter during switch-off phase

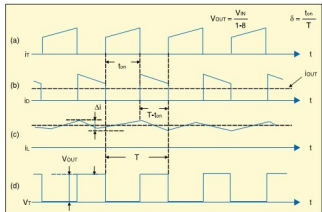


Fig. 3: Current and voltage waveforms during switching: (a) Current through MOSFET, (b) current through Schottky diode, (c) current through the load, and (d) voltage across the MOSFET

a car, you'll need to purchase a vehicle power adaptor separately.

Here we present a DC-to-DC converter design that will allow you to operate your laptop from the 12V battery system of an automobile. It delivers a power of 65 watts, which is sufficient to handle the power requirement of most laptops. This laptop adaptor is basically a boost converter to power from an automobile power point (cigarette lighter socket). Its main features are:

1. Switch-mode power supply
2. Output voltage varying from 16V to 20V
3. Output current of 3 amperes
4. Over-current protection
5. Light weight

Working principle of DC-to-DC converter

The working principle of DC-to-DC converter is simple: an inductor is repeatedly 'charged' and 'discharged' and the resultant induced voltage is added to the supply voltage to produce an output voltage higher than the supply voltage.

Fig. 1 shows the operating principle of the step-up converter.

When the MOSFET conducts, current flows through the coil to produce a magnetic field. The anode of diode D is at ground potential, so it doesn't conduct. The energy previously stored in capacitor C is supplied to the load (R_L).

When the MOSFET stops conducting (refer Fig. 2), current through the coil

no longer has a path to ground through the MOSFET. The magnetic field starts to collapse inducing a voltage across inductor L of the opposite polarity to the switch-on phase. The induced voltage rises above the supply voltage, forward-biasing diode D and transferring energy to capacitor C and load R_L . Energy stored in capacitor C during this phase is transferred to the load during the next switch-on phase. The waveforms shown in Fig. 3 will help you understand the principle easily.

Circuit description

Fig. 4 shows the block diagram of the vehicle power adaptor for laptops. It comprises an over-current protector, pulse-width modulation (PWM) controller, MOSFET switch, inductor, rectifier and filter. The over-current protector turns off the power when the load draws a high current. The PWM

PARTS LIST

Semiconductors:

IC1	- UC3843 current-mode PWM controller
SCR1	- MCR100-6 SCR
T1	- BC547 npn transistor
T2	- IRFZ44 power MOSFET
D1	- 1N4007 rectifier diode
D2	- MBR1645 Schottky diode
LED1	- 5mm bicolor LED

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon unless stated otherwise):

R1, R9	- 470-ohm
R2	- 2.2-kilo-ohm
R3, R5, R6, R11	- 10-kilo-ohm
R4, R10	- 1-kilo-ohm
R7	- 15-kilo-ohm
R8	- 100-kilo-ohm
R12	- 68-kilo-ohm
R13	- 8.2-kilo-ohm
R14	- 100-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 0.01 μ F ceramic disk
C2	- 100 μ F, 35V electrolytic
C3, C4	- 1nF ceramic disk
C5, C6	- 10nF ceramic disk
C7	- 22pF ceramic disk
C8	- 4700 μ F, 50V electrolytic
C9	- 0.1 μ F ceramic disk
C10	- 10 μ F, 35V electrolytic
C11	- 1 μ F, 50V electrolytic

Miscellaneous:

RL1	- 12V, 1C/O relay
S1	- Reed switch
S2	- Push-to-on switch
L1	- 20 μ H inductor coil (see text)
	- 12V DC fan

